

3(a)

one mark each row

cell structure	<i>V. cholerae</i>	epithelial cell from the human intestine
<i>cell wall</i>	✓	✗
<i>cell surface membrane</i>	✓	✓;
<i>ribosomes</i>	✓	✓;
<i>large permanent vacuole</i>	✗	✗;
<i>organelles surrounded by a double membrane</i>	✗	✓;

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Question	Answer	Marks
3(b)	I better sanitation unqualified I <i>ref. to</i> unclean water <i>any five from :</i> <i>ref. to</i> treat drinking water ; A clean / sterilised / chlorinated / safe / bottled / boiled A <i>ref. to</i> water treatment plants provide sewage treatment plants ; A treat waste water / containment of sewage / provide latrines (in temporary camps) ; <i>ref. to</i> keeping sewage and (drinking) water sources separate ; A examples e.g. avoid swimming downstream of sewage outlets avoid defaecating near rivers used for washing water treatment plants upstream of sewage disposal bury faeces <i>ref. to</i> vaccination / providing vaccines ; <i>ref. to</i> education in ways to prevent spread ; <i>ref. to</i> (encourage) hand washing ; <i>ref. to</i> food hygiene ; e.g. use of clean cooking utensils / covering food to prevent flies landing / washing food in treated water / cook food thoroughly / peel fruit and vegetable washed in (suspected) contaminated water discourage use of human faeces for fertiliser / use fertiliser other than human faeces / do not irrigate plant food with contaminated water; control breeding of, vectors / flies ; rapid diagnosis ; rapid treatment (for earlier recovery to minimise risk of spreading) ; <i>ref. to</i> oral rehydration, salts / therapy ; use of antibiotics ; <i>ref. to</i> isolation / quarantine ;	5

Question	Answer	Marks
3(c)(i)	to, bind / attach / AW, HIV / virus / pathogen, to, host cell / T-lymphocyte / T-cell ; (because it) binds / is complementary, to, CD4 receptor proteins / AW ; <i>ref. to binding leads to, entry of virus into cell / endocytosis / fusion of viral envelope with host cell (surface) membrane ;</i>	2
3(c)(ii)	<i>ref. to less cytokine (released) ; in context of fewer helper T-lymphocytes</i> <i>any one from:</i> (so) fewer B-lymphocytes divide by mitosis / AW ; A humoral response decreased (so) fewer B -lymphocytes stimulated to differentiate into plasma cells / AW ; (so) fewer plasma cells to, produce / secrete antibodies ;	2